#5 Cycle through all directions: their indices, in a k-dimensional space, go from `0` to `k-1`.
#6 Check if the variance along the `i`-th axis is larger than the maximum found so far.
#7/#8 If it is so, store the new maximum variance, and update the direction's index.
#9 After the `for` loop cycled through all axis' directions, we can return the index of the direction for which we have found the largest variance.

Of course, the outcome is not always so neat: if the direction of maximum variance is rotated at some angle with respect to the `x` axis (imagine, for instance, the same points rotated 45° clockwise WRT the leaf's centroid), then neither axis direction will produce the optimal result. On average, however, this simpler solution does help.

So, how do we perform the split? We start with listing 10.9, which describes the method to find the direction with maximum variance: it's a simple method performing a global maximum search in a linear space.

Listing 10.10 The SsNode:: updateBoundingEnvelope method

```
function
SsNode::updateBoundingEnvelope()
       #1
  points ←
this.getCentroids()
    #2
  for i in {0..k-1}
do                                                    #3
```